



Technical Service Bulletin

Technical Service Bulletin: TSB180112

Released Date: 05-Jul-2023

New Rear Crankshaft Seal Service Tools

New Rear Crankshaft Seal Service Tools

Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- ISB 4.5 (All Versions)
- ISB 5.9 (All Versions)
- ISB 6.7 (All Versions)
- QSB 4.5 (All Versions)
- QSB 5.9 (All Versions)
- QSB 6.7 (All Versions)
- B4.5G (All Versions)
- B5.9G (All Versions)
- B6.7G (All Versions)
- ISL (All Versions)
- QSL (All Versions)
- ISC (All Versions)
- QSC (All Versions)
- C8.3G (All Versions)

Description of Change

Reusable required service tools to be used for rear crankshaft seal service events have been released.

Reason for Change

Existing service practices can damage adjoining components which can lead to repeat malfunctions.

Customer Communication

Advise self-servicing customers that new required re-usable rear crankshaft seal service tools **must** be used to failsafe installation and removal of rear crankshaft seals in event of leaks moving forward. New required service tools are reusable.

Service Instructions

New required service tools are reusable and meant to replace existing disposable service kits. Associated service literature is being published in affected service manuals.

Centering guide tool **must** be used with front gear train engines.

Service Parts Availability

Service parts are available. See Table 1 for part numbers.

Table 1, Service Parts

Part Description	Existing Part Number	Obsolete	Superseded	New Part Number
Remover, (Unitized) Crankshaft Seal	3164660	-	-	-
Remover, (Lip) Crankshaft Seal	-	-	-	5395330
Installer, (Lip) Crankshaft Seal	-	-	-	5395259
Guide, Centering	-	-	-	5395409
Seal, Oil	5259499	No	Yes	5405392
Kit, Crankshaft Seal	4955566	Yes	Yes	5405392
Kit, Crankshaft Seal	3934486	Yes	Yes	5405392
Kit, Crankshaft Seal	5264562	Yes	Yes	5405392

Publications Affected

Table 2, Publications Affected

Manual Type	Engine	Bulletin Number	Procedure Title	Procedure	Section
Service Manual	B6.7 CM2350 B121B Service Manual	5411223	Crankshaft Seal, Rear	Refer to Procedure 001-024	1
Service Manual	ISB5.9 G CM2180 Service Manual	2883369	Crankshaft Seal, Rear	Refer to Procedure 001-024	1

Table 2, Publications Affected					
Manual Type	Engine	Bulletin Number	Procedure Title	Procedure	Section
Service Manual	QSL9 G CM558 Service Manual	2883375	Crankshaft Seal, Rear Crankshaft Seal Carrier, Rear	Refer to Procedure 001-024 Refer to Procedure 001-104	1
Service Manual	ISB6.7 CM2150 SQ Service Manual	2883504	Crankshaft Seal, Rear	Refer to Procedure 001-024	1
Service Manual	ISB6.7 CM2350 B101 Service Manual	2883567	Crankshaft Seal, Rear	Refer to Procedure 001-024	1
Service Manual	QSB6.7 CM2250 EC (173 HP or Below) Service Manual	2883621	Crankshaft Seal, Rear	Refer to Procedure 001-024	1
Service Manual	C Series Troubleshooting and Repair Manual	3666003	Crankshaft Seal, Rear Crankshaft Seal Carrier, Rear	Refer to Procedure 001-024 Refer to Procedure 001-104	1
Service Manual	B3.9, B4.5, B4.5 RGT, and B5.9 Service Manual	3666087	Crankshaft Seal, Rear	Refer to Procedure 001-024	1

Table 2, Publications Affected					
Manual Type	Engine	Bulletin Number	Procedure Title	Procedure	Section
Service Manual	B5.9G, B5.9LPG, B Gas Plus, B LPG Plus, B Gas International, and G5.9 Troubleshooting and Repair Manual	3666164	Crankshaft Seal, Rear	Refer to Procedure 001-024	1
Service Manual	ISB and QSB5.9-44 Troubleshooting and Repair Manual	3666193	Crankshaft Seal, Rear	Refer to Procedure 001-024	1
Service Manual	C8.3G, C Gas Plus, and L Gas Plus Troubleshooting and Repair Manual	3666206	Crankshaft Seal, Rear Crankshaft Seal Carrier, Rear	Refer to Procedure 001-024 Refer to Procedure 001-104	1
Service Manual	ISB, ISBe, ISBe4, QSB4.5, QSB5.9, and QSB6.7 (Common Rail Fuel System) Service Manual	4021271	Crankshaft Seal, Rear	Refer to Procedure 001-024	1

Table 2, Publications Affected					
Manual Type	Engine	Bulletin Number	Procedure Title	Procedure	Section
Service Manual	QSB3.9-30, QSB4.5-30, and QSB5.9-30 Industrial Troubleshooting and Repair Manual	4021398	Crankshaft Seal, Rear	Refer to Procedure 001-024	1
Service Manual	ISC, ISCe, QSC8.3, ISL, ISLe3, ISLe4 and QSL9 Troubleshooting and Repair Manual	4021418	Crankshaft Seal, Rear Crankshaft Seal Carrier, Rear	Refer to Procedure 001-024 Refer to Procedure 001-104	1
Service Manual	ISC and ISL CM2150 Service Manual	4021569	Crankshaft Seal, Rear Crankshaft Seal Carrier, Rear	Refer to Procedure 001-024 Refer to Procedure 001-104	1
Service Manual	ISB CM2150 and ISB CM2150 E Service Manual	4021578	Crankshaft Seal, Rear	Refer to Procedure 001-024	1
Service Manual	ISLe CM2150 Service Manual	4021630	Crankshaft Seal, Rear Crankshaft Seal Carrier, Rear	Refer to Procedure 001-024 Refer to Procedure 001-104	1
Service Manual	ISL G CM2180 Service Manual	4021649	Crankshaft Seal, Rear Crankshaft Seal Carrier, Rear	Refer to Procedure 001-024 Refer to Procedure 001-104	1

Table 2, Publications Affected					
Manual Type	Engine	Bulletin Number	Procedure Title	Procedure	Section
Service Manual	ISB4.5, ISB6.7, ISD4.5, and ISD6.7 CM2150 SN Service Manual	4022188	Crankshaft Seal, Rear	Refer to Procedure 001-024	1
Service Manual	ISL9 CM2150 SN Service Manual	4022190	Crankshaft Seal, Rear Crankshaft Seal Carrier, Rear	Refer to Procedure 001-024 Refer to Procedure 001-104	1
Service Manual	ISB6.7 CM2250 Service Manual	4022254	Crankshaft Seal, Rear	Refer to Procedure 001-024	1
Service Manual	QSB6.7 CM2250 (174 HP and Above) Service Manual	4022255	Crankshaft Seal, Rear	Refer to Procedure 001-024	1
Service Manual	QSL9 CM2250 Service Manual	4022256	Crankshaft Seal, Rear Crankshaft Seal Carrier, Rear	Refer to Procedure 001-024 Refer to Procedure 001-104	1
Service Manual	ISC8.3 and ISL9 CM2250 Service Manual	4022257	Crankshaft Seal, Rear Crankshaft Seal Carrier, Rear	Refer to Procedure 001-024 Refer to Procedure 001-104	1

Table 2, Publications Affected					
Manual Type	Engine	Bulletin Number	Procedure Title	Procedure	Section
Service Manual	B4.5s and B6.7s Troubleshooting and Repair Manual	4095243	Crankshaft Seal, Rear	Refer to Procedure 001-024	1
Service Manual	Marine QSB6.7 CM2250 Service Manual	4310611	Crankshaft Seal, Rear	Refer to Procedure 001-024	1
Service Manual	Marine Auxiliary QSB7-DM CM850 Service Manual	4310679	Crankshaft Seal, Rear	Refer to Procedure 001-024	1
Service Manual	L8.9 Series Troubleshooting and Repair Manual	4310704	Crankshaft Seal, Rear Crankshaft Seal Carrier, Rear	Refer to Procedure 001-024 Refer to Procedure 001-104	1
Service Manual	ISBe and ISDe CM2150 Service Manual	4021597	Crankshaft Seal, Rear	Refer to Procedure 001-024	1

--

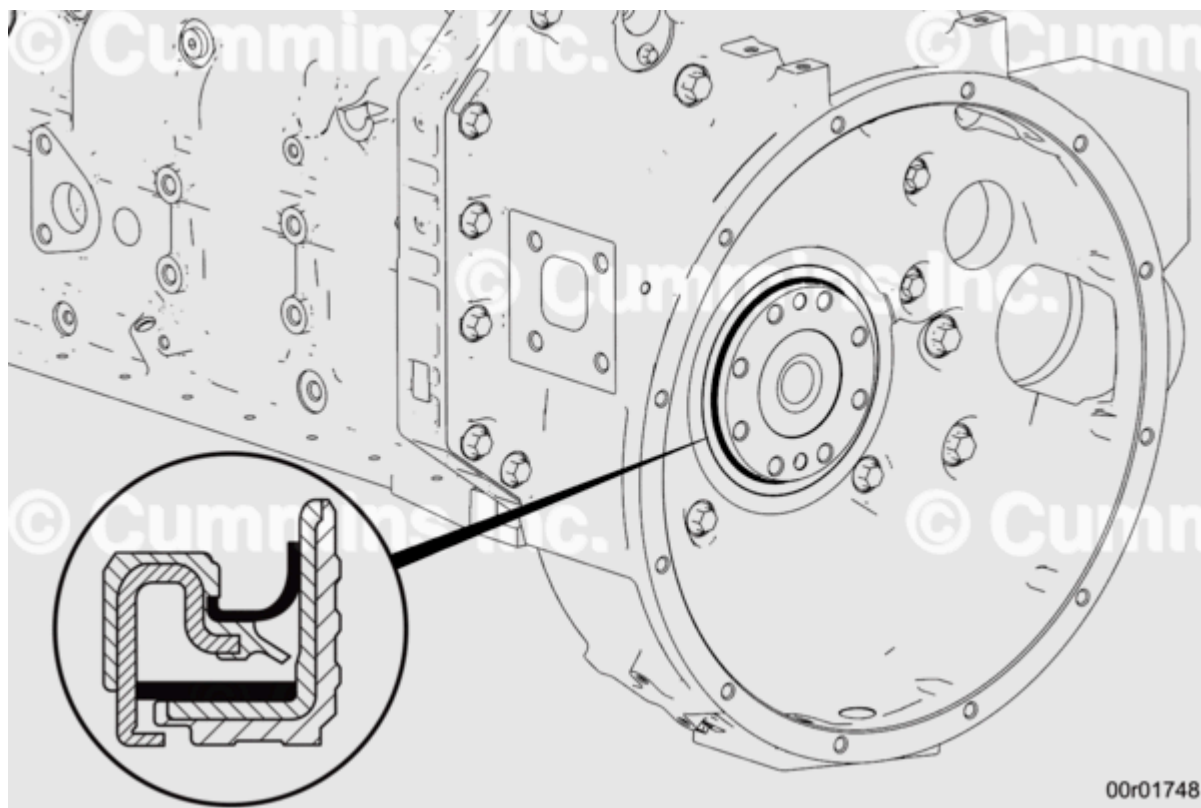


Figure 1, Unitized Crankshaft Seal

Figure 1 shows an example of a unitized crankshaft seal.

Some engines used to use a dual or non-lip style seal which utilize a built in wear sleeve and a concealed sealing lip. The inner and outer diameters are press-fit onto the crankshaft and the flywheel housing respectively. The seal is removed from the flywheel housing/seal carrier bore using required service tools, rear crankshaft seal removal kit, Part Number 3164660.

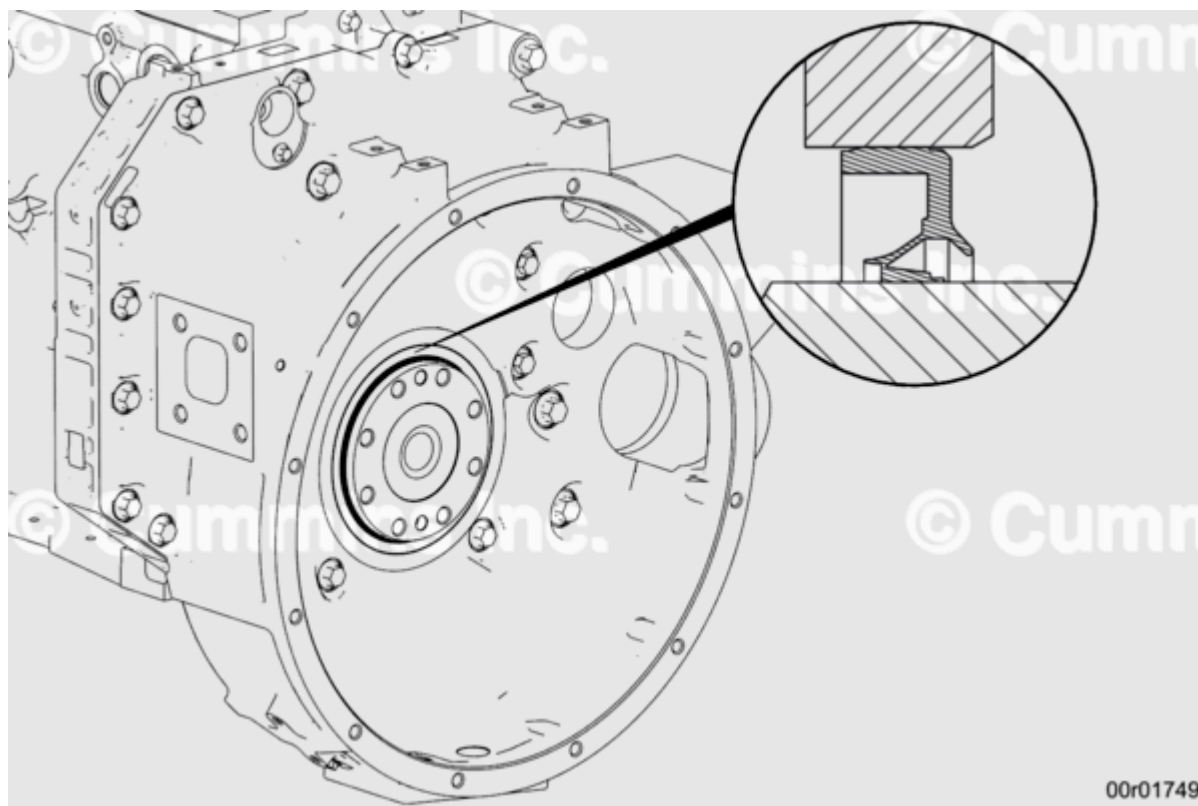


Figure 2, Lip Crankshaft Seal

Figure 2 shows an example of a lip crankshaft seal.

The rotating portion of the sealing occurs at the contact surface between the lip of the seal and the crankshaft. The seal is installed into the flywheel housing/seal carrier bore using required service tools, rear crankshaft seal installation kit, Part Number 5395259, and rear crankshaft seal removal kit, Part Number 5395330.

Document History

Date	Details
2018-8-16	Module Created
2018-9-24	Updated Table 1
2023-6-30	Added information on unitized seal description vs lip seal.
2023-7-5	Minor wording correction.

Last Modified: 05-Jul-2023